CONFIDENTIAL — CUSTOMER DELIVERABLE

# Installation and Configuration Record

Post-Deployment As-Built Package — azlocal-iic-01

Mode: as-built | Ranger Version: 2.1.0 | Generated: 04/06/2026 12:08:32

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## Priority Recommendations

- **[WARNING]** One or more node certificates expire within 90 days
  Review certificate ownership and renew expiring node certificates before handoff.
- **[WARNING]** iDRAC firmware below recommended baseline on all nodes
  Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- **[INFORMATIONAL]** Azure Policy assignments discovered at resource group scope
  Verify policy scope extends to arc resource group and monitor for compliance drift.
- **[INFORMATIONAL]** Cluster event history includes a transient network quorum warning
  Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

## Collector Overview

### hardware

**Status:** success

**Targets:**

### management-performance

**Status:** success

**Targets:**

### monitoring-observability

**Status:** success

**Targets:**

### storage-networking

**Status:** success

**Targets:**

### topology-cluster

**Status:** success

**Targets:**

### waf-assessment

**Status:** success

**Targets:**

### workload-identity-azure

**Status:** success

**Targets:**

## Document Control

| | |
|---|---|
| Document Title | Azure Local As-Built Documentation — azlocal-iic-01 |
| Package ID | azlocal-iic-01-as-built-20260417T020635Z |
| Report Tier | management |
| Revision | 1.0 (initial handoff) |
| Classification | CONFIDENTIAL — CUSTOMER DELIVERABLE |
| Prepared By Azure Local Ranger v2.1.0 | |
| Prepared On | 04/06/2026 12:08:32 |
| Schema Version | 1.2.0-draft |
| Document Status | FINAL — AS-BUILT HANDOFF |

## Run Summary

- Mode: as-built
- Generated: 04/06/2026 12:08:32
- Cluster: azlocal-iic-01
- Nodes: 3
- Collectors: 7/7 successful
- Artifacts currently recorded: 0

## Readiness Snapshot

- Schema validation: passed
- Critical findings: 0
- Warning findings: 2
- Informational findings: 2
- Successful collectors: 7 of 7
- Partial or failed collectors: 0

## Environment Overview

- VMs discovered: 5
- Azure resources discovered: 10
- Connected Azure auth method: service-principal
- Monitoring resources: 5
- Management services running: 4

## Topology and Operating Model

- Deployment type: hyperconverged
- Identity mode: ad
- Connectivity mode: connected
- Storage architecture: storage-spaces-direct
- Network architecture: switched
- Variant markers: connected

## Domain Coverage

- Cluster nodes: 3
- Cluster roles: 5
- Storage pools: 1
- Physical disks: 24
- Network adapters: 12
- VMs discovered: 5
- Azure resources: 10
- Alerting resources: 2
- Management services: 4

## Node Inventory

| Node | Model | State | OS | OS Version / Build | Logical CPUs | RAM (GiB) |
|---|---|---|---|---|---|---|
| azl-iic-n01 | PowerEdge R760 | Up | Microsoft Azure Stack HCI | 10.0.25398.1189 | 64 | 512 |
| azl-iic-n02 | PowerEdge R760 | Up | Microsoft Azure Stack HCI | 10.0.25398.1189 | 64 | 512 |
| azl-iic-n03 | PowerEdge R760 | Up | Microsoft Azure Stack HCI | 10.0.25398.1189 | 64 | 512 |

## Collector Status

- hardware: success ()
- management-performance: success ()
- monitoring-observability: success ()
- storage-networking: success ()
- topology-cluster: success ()
- waf-assessment: success ()
- workload-identity-azure: success ()

## Operational Risk Summary

- Nodes not healthy: 0
- Unhealthy disks: 0
- High CPU nodes: 0
- Certificates expiring within 90 days: 1
- Azure Policy assignments: 2
- Schema warnings: 0
- Azure Advisor recommendations: total, 3 high-impact

## Workload Summary

- Total VMs: 5
- Total nodes: 3
- AKS clusters: 1
- Arc-connected machines: 1
- Update compliance: No update resources tracked
- Licensing: IIC Platform Production
- VMs using Arc IP fallback: Not collected

## Capacity Summary

- Storage total raw: 0 TiB
- Storage total usable: 0 TiB
- Storage used by workloads: 0 TiB (0% of usable)
- Storage reserve target: 0 TiB (0% of usable)
- Storage free usable: 0 TiB
- Projected safe allocatable: 0 TiB
- Thin provisioning ratio: Not computed
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Average VMs per node: N/A

## Priority Recommendations

- [WARNING] One or more node certificates expire within 90 days: Review certificate ownership and renew expiring node certificates before handoff.
- [WARNING] iDRAC firmware below recommended baseline on all nodes: Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.
- [INFORMATIONAL] Azure Policy assignments discovered at resource group scope: Verify policy scope extends to arc resource group and monitor for compliance drift.
- [INFORMATIONAL] Cluster event history includes a transient network quorum warning: Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.

## VM Inventory

| Name | State | vCPU | RAM (GiB) | Host Node | Generation |
|---|---|---|---|---|---|
| avd-iic-sh01 | Running | 8 | 16 | azl-iic-n01 | 2 |
| avd-iic-sh02 | Running | 8 | 16 | azl-iic-n02 | 2 |
| avd-iic-sh03 | Running | 8 | 16 | azl-iic-n03 | 2 |
| arc-iic-vm01 | Running | 4 | 8 | azl-iic-n01 | 2 |
| arc-iic-vm02 | Running | 4 | 8 | azl-iic-n02 | 2 |

## VM Density Metrics

- VMs per node (average): N/A
- Highest-density node: N/A
- vCPU:pCPU overcommit ratio: Not computed
- Memory overcommit ratio: Not computed
- Arc-connected VMs:
- VMs using Arc IP fallback:
- Avg CPU utilization (all nodes): 38.3%
- Avg available memory (all nodes): 176 GiB

## Storage Pool Capacity

| Pool | Raw (GiB) | Usable (GiB) | Used (GiB) | Reserve (GiB) | Safe Alloc (GiB) | Posture |
|---|---|---|---|---|---|---|
| — | 0 | 0 | 0 | 0 | 0 | — |

## ESU Enrollment

- Eligible Arc-connected VMs:
- Enrolled in ESU:
- Not enrolled in ESU:
- Ineligible:
-

## Coverage Assessment

- Monitoring coverage (AMA): 100% (3 of 3 nodes have Azure Monitor Agent)
- Backup coverage estimate: 20% (based on backup items vs VM count)
- Defender for Cloud: Not confirmed or not collected
- WDAC policy: Not collected
- BitLocker status: Not collected
- Certificates expiring in <90 days: 1

## Security Posture Summary

- Secured-Core enabled nodes: of 3
- Syslog forwarding nodes:
- RBAC assignments at resource group: 1
- Policy assignments tracked: 2
- Policy exemptions:
- Active Directory site: Not collected

## Management Tool Coverage

- WAC installed nodes: of 3
- SCVMM agent nodes:
- SCOM agent nodes:
- Running management services: 4
- Third-party agent types: (0)

## Cost & Licensing

- AHB status: Enabled
- Total physical cores: 96 | AHB-covered: 96 | Unenrolled: 0
- Current monthly cost: 960 USD @ 10/core/month
- AHB adoption: 100%
- Potential monthly savings (if remaining cores enrolled in AHB): 0 USD
- Pricing based on Azure Local public pricing (10 USD/physical core/month) as of 2026-04-16.
- For current rates, see: https://azure.microsoft.com/en-us/pricing/details/azure-local/

## Cost & Licensing — Per Node

| Node | Physical Cores | AHB Enabled | Monthly Cost (USD) | Monthly Saving (USD) |
|---|---|---|---|---|
| azl-iic-n01 | 32 | Yes | 320 | 320 |
| azl-iic-n02 | 32 | Yes | 320 | 320 |
| azl-iic-n03 | 32 | Yes | 320 | 320 |

*Pricing as of 2026-04-16 — https://azure.microsoft.com/en-us/pricing/details/azure-local/*

## Arc Extensions by Node

| Node | Name | Type | Publisher | Version | State |
|---|---|---|---|---|---|
| azl-iic-n01 | AzureMonitorWindowsAgent | AzureMonitorWindowsAgent | Microsoft.Azure.Monitor | 1.22.2.0 | Succeeded |
| azl-iic-n01 | GuestConfigExtension | ConfigurationforWindows | Microsoft.GuestConfiguration | 1.29.78.0 | Succeeded |
| azl-iic-n01 | MicrosoftDefenderForServers | MDE.Windows | Microsoft.Azure.AzureDefenderForServers | 1.0 | Succeeded |
| azl-iic-n02 | AzureMonitorWindowsAgent | AzureMonitorWindowsAgent | Microsoft.Azure.Monitor | 1.22.2.0 | Succeeded |
| azl-iic-n02 | GuestConfigExtension | ConfigurationforWindows | Microsoft.GuestConfiguration | 1.29.78.0 | Succeeded |
| azl-iic-n02 | MicrosoftDefenderForServers | MDE.Windows | Microsoft.Azure.AzureDefenderForServers | 1.0 | Succeeded |
| azl-iic-n03 | AzureMonitorWindowsAgent | AzureMonitorWindowsAgent | Microsoft.Azure.Monitor | 1.22.2.0 | Succeeded |
| azl-iic-n03 | GuestConfigExtension | ConfigurationforWindows | Microsoft.GuestConfiguration | 1.29.78.0 | Succeeded |
| azl-iic-n03 | MicrosoftDefenderForServers | MDE.Windows | Microsoft.Azure.AzureDefenderForServers | 1.0 | Succeeded |

*AMA coverage: 100% of nodes. Failed extensions: 0.*

## Logical Networks

| Name | VM Switch | DHCP | Subnets | State |
|---|---|---|---|---|
| lnet-iic-mgmt-01 | ConvergedSwitch | No | 1 | Succeeded |
| lnet-iic-workload-01 | ConvergedSwitch | Yes | 1 | Succeeded |

## Logical Network Subnets

| Network | Subnet | Address Prefix | VLAN | IP Pools |
|---|---|---|---|---|
| lnet-iic-mgmt-01 | subnet-management | 10.0.0.0/24 | 2203 | 1 |
| lnet-iic-workload-01 | subnet-workload | 10.0.2.0/24 | 2205 | 1 |

## Storage Paths

| Name | Path | Available (GB) | File System | State |
|---|---|---|---|---|
| sp-iic-vmstore-01 | C:\ClusterStorage\csv-iic-azlocal-01-vmstore-01\ArcVmDisks | 1480 | CSVFS_ReFS | Succeeded |
| sp-iic-vmstore-02 | C:\ClusterStorage\csv-iic-azlocal-01-vmstore-02\ArcVmDisks | 1620 | CSVFS_ReFS | Succeeded |
| sp-iic-vmstore-03 | C:\ClusterStorage\csv-iic-azlocal-01-vmstore-03\ArcVmDisks | 1710 | CSVFS_ReFS | Succeeded |

## Arc Resource Bridge

| Name | Status | Version | Distro | Provisioning |
|---|---|---|---|---|
| rb-iic-azlocal-01 | | | | |
| Connected | | | | |
| 1.5.23 | | | | |
| MOC-K8S | | | | |
| Succeeded | | | | |

## Custom Locations

| Name | Namespace | Location | State |
|---|---|---|---|
| cl-iic-azlocal-01 | | | |
| azlocal | | | |
| eastus | | | |
| Succeeded | | | |

## Arc Gateways

| Name | Endpoint | Allowed Features | State |
|---|---|---|---|
| arcgw-iic-01 | | | |
| https://arcgw-iic-01.gw.arc.azure.com | | | |
| * | | | |
| Succeeded | | | |

## Marketplace & Custom Images

| Name | Type | OS | Version | Size (GB) | State |
|---|---|---|---|---|---|
| img-iic-ws2022-datacenter | Marketplace | Windows | 20348.2700.240906 | 127 | Succeeded |
| img-iic-rhel9-base | Custom | Linux | 9.4.1 | 24 | Succeeded |

## Arc Agent Versions

| Version | Node Count | Nodes |
|---|---|---|
| 1.39.02750.1478 | 2 | azl-iic-n01, azl-iic-n02 |
| 1.38.02620.1234 | 1 | azl-iic-n03 |

*Drift status: warning. Latest: 1.39.02750.1478. Max behind: 1.*

## VM Distribution by Node

| Node | VM Count |
|---|---|
| azl-iic-n01 | 2 |
| azl-iic-n02 | 2 |
| azl-iic-n03 | 1 |

*Balance: balanced | CV: 0.283 | Status: warning*

## Installation Register (Bill of Materials)

| Hostname | FQDN | Manufacturer | Model | Serial | BIOS at Deployment | OS Installed | OS Build |
|---|---|---|---|---|---|---|---|
| azl-iic-n01 | azl-iic-n01.iic.local | Dell Inc. | PowerEdge R760 | Not recorded | 2.10.0.0 | Microsoft Azure Stack HCI | 10.0.25398.1189 |
| azl-iic-n02 | azl-iic-n02.iic.local | Dell Inc. | PowerEdge R760 | Not recorded | 2.10.0.0 | Microsoft Azure Stack HCI | 10.0.25398.1189 |
| azl-iic-n03 | azl-iic-n03.iic.local | Dell Inc. | PowerEdge R760 | Not recorded | 2.10.0.0 | Microsoft Azure Stack HCI | 10.0.25398.1189 |

*Each unit listed above was installed and commissioned as part of this deployment. Serial numbers are recorded as discovered at handoff; missing values indicate the collector could not access the field.*

## Per-Node Configuration Record

| Node | State at Handoff | Logical CPUs | Installed Memory | Domain Joined | BIOS Version |
|---|---|---|---|---|---|
| azl-iic-n01 | Up | 64 | 512 GiB | iic.local | 2.10.0.0 |
| azl-iic-n02 | Up | 64 | 512 GiB | iic.local | 2.10.0.0 |
| azl-iic-n03 | Up | 64 | 512 GiB | iic.local | 2.10.0.0 |

*Each node was configured with the values above at the time of deployment.*

## Network Address Allocation Record

| Cluster Network | Role | Network Address | Mask | Metric | State |
|---|---|---|---|---|---|
| Management | Cluster + Client (Management) | 10.0.0.0 | 255.255.255.0 | 100 | Up |
| Storage-1 | Cluster Only (Storage) | 10.0.1.0 | 255.255.255.0 | 200 | Up |
| Storage-2 | Cluster Only (Storage) | 10.0.1.128 | 255.255.255.128 | 200 | Up |
| Workload | Cluster + Client (Workload) | 10.0.2.0 | 255.255.255.0 | 300 | Up |

*Cluster networks were assigned and configured as recorded above during deployment.*

## Storage Configuration Record

| Storage Pool | Raw Capacity | Usable Capacity | Health at Handoff |
|---|---|---|---|
| S2D on azlocal-iic-01 | 43068 GiB | — | Healthy |

*Storage pools and their capacities were provisioned at deployment as shown. CSV and virtual-disk details are included in the delivery-registers workbook.*

## Identity and Security Record

| | |
|---|---|
| Identity mode | ad |
| Active Directory site | Not collected |
| Secured-Core nodes enrolled | of 3 |
| BitLocker | Not collected |
| WDAC policy | Not collected |
| Certificates tracked | 1 |
| Certificates expiring <90d | 1 |
| RBAC assignments at RG scope | 1 |

## Azure Integration Record

| | |
|---|---|
| Tenant ID | 00000000-0000-0000-0000-000000000000 |
| Subscription ID | 33333333-3333-3333-3333-333333333333 |
| Resource Group | rg-iic-compute-01 |
| Arc-connected machines | 1 |
| AKS clusters | 1 |
| Azure Monitor Agents | 3 |
| Backup items | 1 |
| ASR protected items | 1 |

## Validation Record

| | |
|---|---|
| Validation report | See cluster-validation report artifact (Test-Cluster output) |
| Collectors run | 7 |
| Collectors successful | 7 |
| Collectors partial | 0 |
| Collectors failed | 0 |
| Schema validation | Passed |
| Critical findings | 0 |
| Warning findings | 2 |

## Known Issues and Deviations

| Severity | Item | Deviation | Remediation Path |
|---|---|---|---|
| WARNING | One or more node certificates expire within 90 days | Identity posture data indicates certificate on azl-iic-n01 expires within 90 days. | Review certificate ownership and renew expiring node certificates before handoff. |
| WARNING | iDRAC firmware below recommended baseline on all nodes | OEM integration data shows iDRAC firmware at 7.10.30.00. The recommended baseline is 7.20 or later. | Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window. |

*Deviations listed below were documented at handoff. Items are accepted as-built unless explicitly marked for follow-up remediation.*

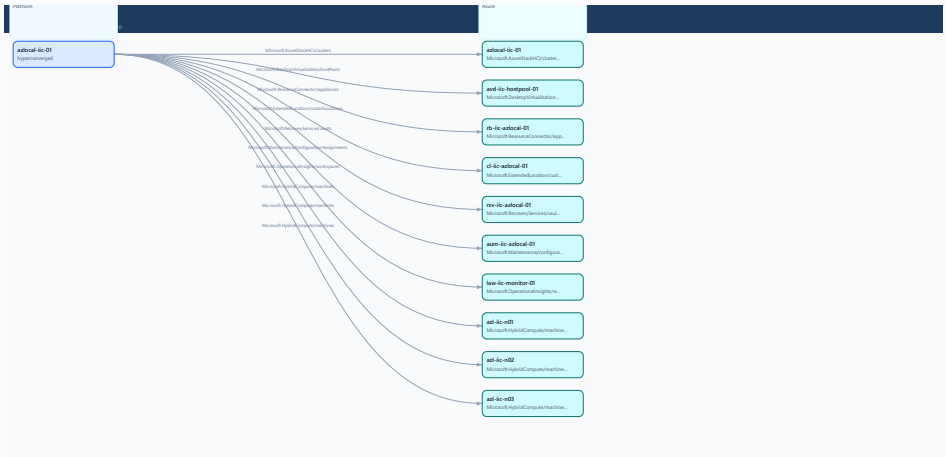
## Acceptance and Sign-Off

This as-built package was generated from the Ranger discovery run on the date shown in the Document Control section. Review all findings before accepting this package as the formal handoff record.

| Role | Name | Date | Signature |
|---|---|---|---|
| Implementation Engineer | | | |
| Technical Reviewer | | | |
| Customer Representative | | | |

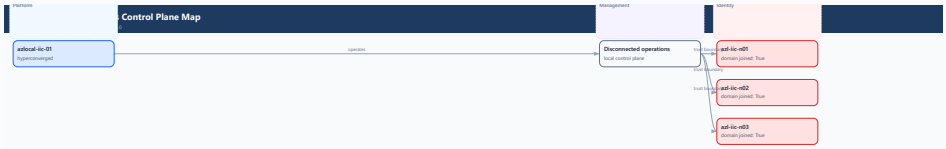
# Architecture Diagrams

## Azure Arc Integration




## Backup Recovery Map

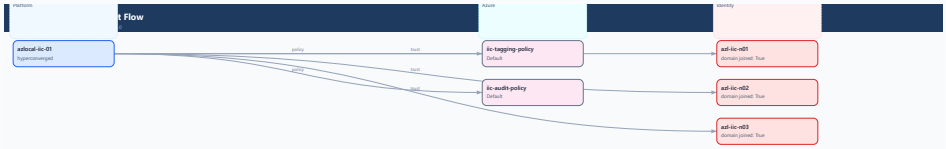
## Disconnected Control Plane



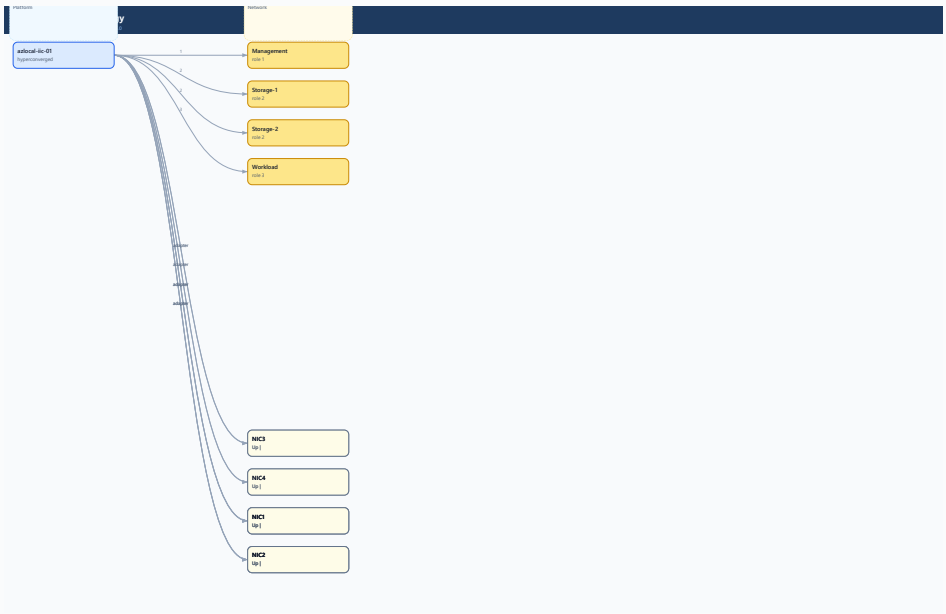
## Fabric Map

## Identity Access Surface

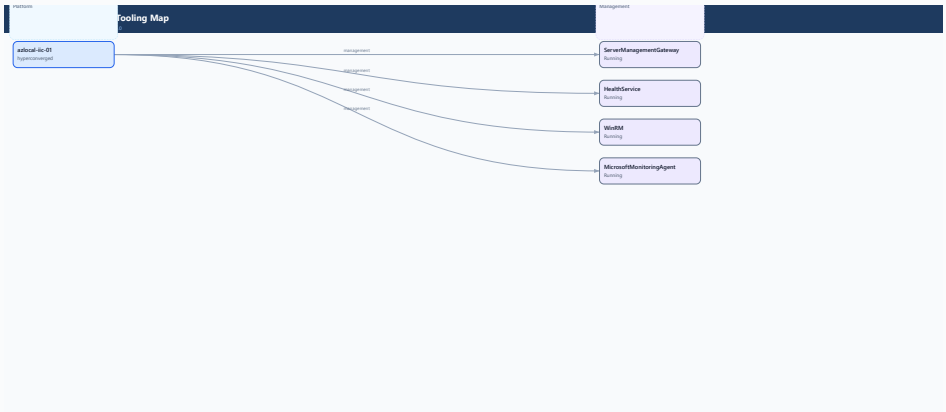
## Identity Secret Flow



## Logical Network Topology

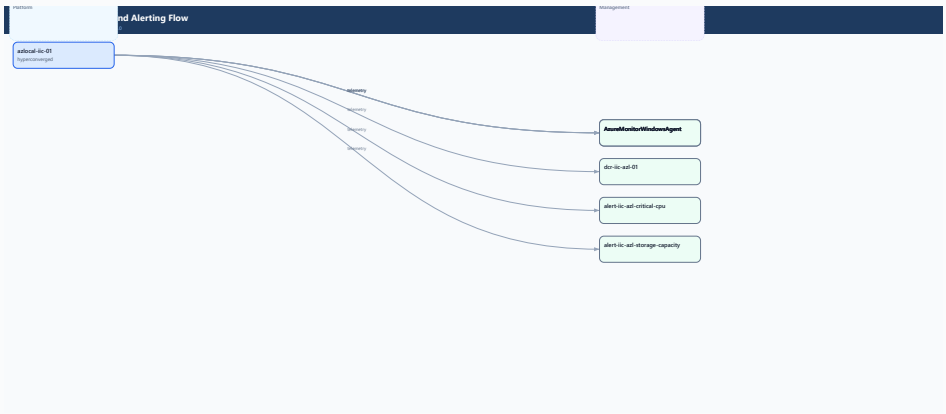



## Management Plane Tooling



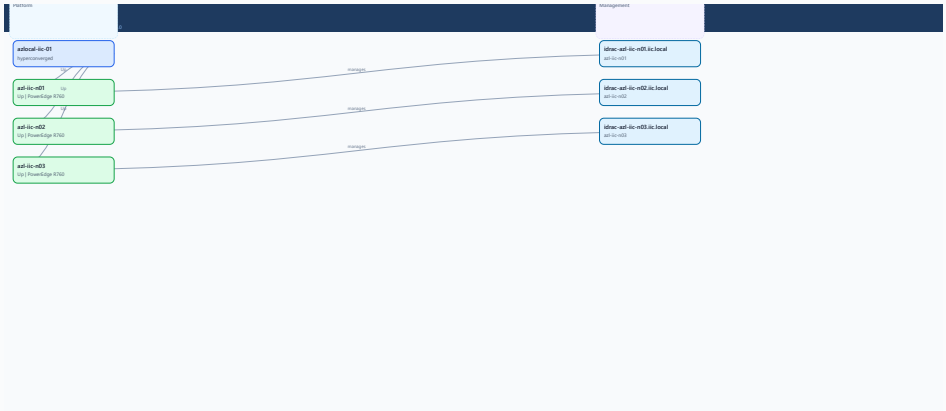

## Monitoring Health Heatmap

## Monitoring Telemetry Flow



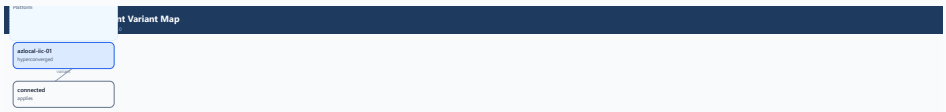

## Oem Firmware Posture

## Physical Architecture



## Storage Architecture

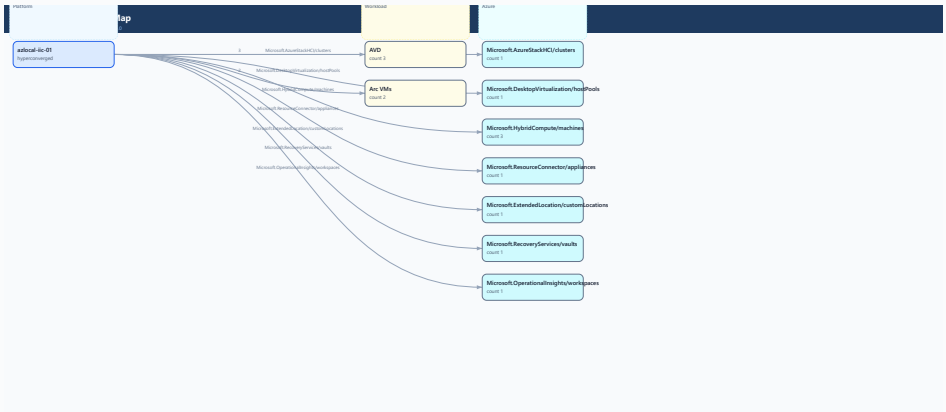
## Topology Variant Map




## Vm Placement Map

## Workload Family Placement

## Workload Services Map



## Findings

WARNING **One or more node certificates expire within 90 days**

Identity posture data indicates certificate on azl-iic-n01 expires within 90 days.

**Current state:** 1 certificate expiring within 90 days

**Recommendation:** Review certificate ownership and renew expiring node certificates before handoff.

WARNING **iDRAC firmware below recommended baseline on all nodes**

OEM integration data shows iDRAC firmware at 7.10.30.00. The recommended baseline is 7.20 or later.

**Current state:** iDRAC firmware 7.10.30.00

**Recommendation:** Update iDRAC firmware via Dell OME or Lifecycle Controller during next maintenance window.

INFORMATIONAL **Azure Policy assignments discovered at resource group scope**

2 policy assignments are enforcing tagging and security control auditing.

**Current state:** 2 policy assignments active

**Recommendation:** Verify policy scope extends to arc resource group and monitor for compliance drift.

INFORMATIONAL **Cluster event history includes a transient network quorum warning**

A Management network quorum warning event was recorded on 2026-04-06. Cluster recovered automatically.

**Current state:** single event logged; cluster healthy

**Recommendation:** Review switch port configuration to confirm management NICs are on dedicated VLANs and bonded correctly.